

# Zhuoran Li

+86 151-8984-3657 | airplane@sjtu.edu.cn | github.com/iairplane

## RESEARCH INTERESTS

Reliable vision-language-action models; embodied AI and robot learning; failure diagnosis and recovery; memory-augmented agents; long-horizon manipulation; multimodal and learning systems.

## EDUCATION

Shanghai Jiao Tong University (SJTU), Shanghai, China Sep. 2023 – Jun. 2027  
(Expected)

*B.Eng. in Software Engineering, School of Computer Science*

GPA: **92.2/100**; Rank: **3/95**. Undergraduate researcher at the Institute of Parallel and Distributed Systems (IPADS), advised by Prof. Jinyu Gu.

## SELECTED PUBLICATIONS

- [1] **Zhuoran Li\***, Zhiyang Li\*, Kaijun Zhou, and Jinyu Gu. “RoboHarness: A Memory-Augmented Policy Harness for Vision-Language-Action Model Robustness via In-Context Adaptation.” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. **Accepted**. \*Equal contribution; Zhuoran Li listed first.
- [2] Yucen Gao, Zhemeng Yu, **Zhuoran Li**, Jianxiong Guo, and Xiaofeng Gao. “GKD-Recruiter: Jointly Modeling Social and Task Heterogeneity for Spatial Crowdsourcing via Graph Knowledge Distillation.” *International Conference on Machine Learning (ICML)*, 2026. **Accepted as a regular paper**.
- [3] **Zhuoran Li**, Yucen Gao, Yu Yin, Xinle Li, Hui Gao, Xiaofeng Gao, and Guihai Chen. “DRNCS: Dual-Level Route Generation Model Based on Node Contraction and Shortcuts.” *European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)*, 2025. **Published; sole first author**.
- [4] Yucen Gao, **Zhuoran Li**, Jingyu He, Jun Fang, Hui Gao, Xiaofeng Gao, and Guihai Chen. “Algorithms for Shortest Path Tour Problem.” *Theoretical Computer Science*, vol. 1054, art. 115428, 2025. **Published**.

## RESEARCH EXPERIENCE

**RoboHarness: Memory-Augmented Policy Harness for VLA Robustness** Dec. 2025 – Jun. 2026

*Co-first Author (first-listed) | Accepted at IROS 2026*

- Designed an inference-time reliability harness around frozen VLA policies, combining contrastive Dual-Memory retrieval, attribution-driven vision-language orchestration, MCP-based tool interventions, and offline memory consolidation.
- Led framework design, core implementation, experiments, and manuscript preparation; evaluated  $\pi_0$ ,  $\pi_{0.5}$ , and SmolVLA on LIBERO-PRO and LIBERO-RoboHarness without parameter fine-tuning, yielding a 56.6-percentage-point average absolute success-rate gain and up to 89.1 percentage points on long-horizon tasks.

**ManiLadder: Multi-Level Benchmark for Manipulation Intelligence** Sep. 2025 – Present  
*Core Developer and Manuscript Contributor | Manuscript in preparation*

- Built a modular simulation and data pipeline spanning task construction, demonstration collection, policy training, and automated evaluation; the benchmark contains **114 tasks**, **5,700 human demonstrations**, and four difficulty levels.
- Contributed to the benchmark’s simulation environments, reproducible evaluation workflow, and manuscript, supporting unified comparison of imitation-learning and VLA policies across task levels.

## RESEARCH EXPERIENCE (CONTINUED)

---

**GKD-Recruiter: Graph Knowledge Distillation for Spatial Crowdsourcing** Jan. 2025 – Jun. 2026

*Researcher | ICML 2026 regular paper accepted*

- Studied spatial task allocation with heterogeneous task requirements and worker attributes through task-worker graph representations and graph knowledge distillation.
- Contributed to algorithm modeling, experimental analysis, and manuscript preparation, with an emphasis on learning task-worker matching relations at scale.

**DRNCS: Dual-Level Route Generation via Node Contraction and Shortcuts** Jun. 2024 – Oct. 2024

*Sole First Author | Published at ECML PKDD 2025*

- Proposed shortcut-edge differential contraction and dynamic probabilistic shortcut injection to compress dense road-network search spaces; achieved **0.3–0.8 ms/trip** latency and **3,356 trips/s** throughput.
- Independently carried out method design, system implementation, experimental validation, and manuscript writing from problem formulation to publication.

**Algorithms for the Shortest Path Tour Problem (SPTP)** Nov. 2024 – Mar. 2026

*Algorithm Researcher and Co-Author | Published in Theoretical Computer Science*

- Contributed to problem formalization, Stage and Global Dijkstra algorithms, and complexity analysis, achieving  $O(\ell(n + m) \log n)$  time and  $O(\ell n)$  space for exact SPTP search.
- Evaluated the algorithms on million-node road networks and obtained up to a **50× speedup** over baseline methods.

## RESEARCH APPOINTMENTS AND INTERNATIONAL EXPERIENCE

---

**Institute of Parallel and Distributed Systems (IPADS), SJTU** Jun. 2025 – Present

*Undergraduate Research Intern; Advisor: Prof. Jinyu Gu*

**LinS Lab, National University of Singapore (NUS)** Summer 2025

*Summer Research Intern*

**Imperial College London** Winter 2025

*Winter School in Data Science and Artificial Intelligence*

## ONGOING RESEARCH

---

- Contributing to the system-software taxonomy, literature review, and manuscript preparation for *Beyond the Model: A Survey of System Software for Embodied Agents* (2026–Present).
- Helped organize the literature around planning, skill and tool management, execution control, memory and world state, verification and recovery, adaptation, and multi-agent coordination.

## SELECTED HONORS AND AWARDS

---

Chun-Tsung Foundation Award (2024)

Zhiyuan Honor Scholarship (2023–2025); Academic Scholarship (2024–2025)

First Prize, 16th Shanghai Undergraduate Mathematics Competition, Non-Mathematics Category A (2024)

Jiangsu Province Excellent Student Award (2023)

## TECHNICAL SKILLS

---

|                          |                                                                                                                     |
|--------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>       | C++, Python, Java, JavaScript/TypeScript                                                                            |
| <b>AI and Robotics</b>   | PyTorch; computer vision; reinforcement learning; graph learning; imitation learning; VLA and multimodal evaluation |
| <b>Systems and Tools</b> | Linux, Git, LaTeX, React, Spring Boot, Nginx, Redis                                                                 |
| <b>English</b>           | CET-4: 624; CET-6: 605                                                                                              |